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Case Studies

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Forecasting Electrical Power Consumption Using Individual Load Profiles

Supervisors:

Prof. Dr. Christine Müller

Dr. Mirko Jakubzik

Author: Najam Ul Hasan

Group Members:

Akash Shambhunath Kanojiya

Theja Sanduni Mahabalage Don

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1 Introduction

The transformation of the energy sector, driven by the increased integration of renewable resources and the electrification of heating and transport, has made understanding the household electricity consumption patterns a crucial topic. In particular, the residential heat pumps are becoming a mainstream sustainable alternative to the conventional fossil fuel-based heating systems. However, their adoption also introduces significant complexity and variability to household demand profiles, with implications for grid stability, demand forecasting, and the formulation of demand-side management strategies. (Fischer and Madani, 2017, Sajjad et al., 2014)

The project aims to contribute to the understanding of electricity consumption of households equipped with heat pumps, with an idea of projecting forecasted electricity consumption using the prior available data. Our approach to the objective involves data transformation, advanced statistical modeling, and comparative analysis. We start with the preprocessing and smoothing of quarter-hour consumption data, realizing the intra- and inter-day patterns. Multivariate statistical models, including MANOVA and Multivariate Linear Regression, to evaluate the choice of model and scaling of the resulting OLS coefficients, which are the key component for clustering the households and provide an insight to how different households possess respective load profiles. We then calculate the Mean Load Profile for clusters and compare for each smoothing method, as these will be vital for the comparative forecasting.

The following report is comprised of sections: Section 2 details the objective of the project and explains the data used for the analysis. In Section 3, the statistical methods used for the project are highlighted, including their mathematical formulations, and the software used to implement them. Following that in Section 4, a thorough statistical analysis is presented with the interpretation of results and graphical summaries. Lastly, Section 5 concludes the report with a summary of key findings, discusses the implications and provides an outlook on possible further research directions.

2 Problem and Data Description

2.1 Research Hypotheses

Do the households change their consumption pattern on the weekends compared to weekdays? Which of the forecasting methods provides better results for the households? Do we observe more accurate forecasts using the mean load profiles, or is only the recorded data sufficient to forecast?

In this project, we conduct various forecasts for three randomly chosen households using advanced state-space models, namely Kalman Filter (Kalman, 1960) and Particle Filter (Gordon et al., 1993). For each household, we initially performed the forecast simply with the available recorded data for each hour, and incorporated the mean load profiles separately for the comparative predictions. The following methods helped to capture the future values using the past values and the respective noise, therefore observing a pattern of projected consumption compared to the electricity originally consumed. For the forecast methods, the performance was evaluated using the root mean squared error (RMSE). Refer to Section 3 for a detailed explanation of the statistical methods.

2.2 Data Material

In the following section we describe the variables used to develop our analysis. The data was taken from the paper "Dataset on electrical single-family house and heat pump load profiles in Germany" (Schlemminger et al., 2022). The sources provides crucial data to understand the consumption of electricity for the selected households within Northern Germany.

For the particular project, we had the data for 33 households, each as a separate database starting from January 1st, 2019 to December 31st, 2019. Each household was contained in their separate database as a CSV file, and contained the electricity consumed (measured in Watts) in the "TOTAL_LOAD" column, recorded at every 15-minute interval for the time period. Each household dataset had approximately 34,800 data points, not accounting for the missing points that are to be mentioned.

Like any other data source, we ran into the issue of missing values. These could be described as of two types: one of the types of missing data being the missing days, that is there may be no data available for the whole day in a household. The other type of missing data being

the missing intermittent points, that is, some data may not be available within, such as the missing 15-min interval between two neighboring dataset. For the missing days, there were six households with the missing days in the following table:

Household	Missing Days
SFH17	122
SFH25	76
SFH31	31
SFH37	208
SFH40	109
SFH6	282

Table 1: Missing Days per Household

As there were households with the consumption data missing for more than half of the year in the study, we kept the missing dates as-is, however these could have implications with clustering, and forecasting as described further in the report.

2.2.1 Data Transformation

As an initial step, we first explain the transformations applied to the data. We started with the smoothing of the datasets by calculating moving averages for the hourly data for each household. Two forms of moving averages were used for smoothing, namely Hourly Moving Average (Ya) and the Weekly Moving Average (Yb). For the former, we set the window for ± 30 minutes for the smoothing and in the case of Weekly Moving Average, the averaging window was set to ± 2 weeks. For comparison, here are the summary statistics for Household 12, with the original data and the smoothed counterparts:

Min	1st Qu.	Median	Mean	3rd Qu.	Max
37.45	123.54	162.70	329.90	245.98	6457.23

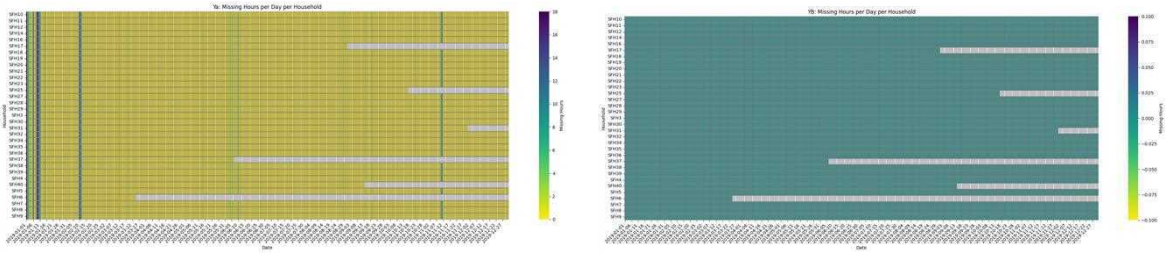
Table 2: SFH12 Raw Data Summary Statistics

	Min	1st Qu.	Median	Mean	3rd Qu.	Max
Ya	112.4	229.6	327.4	536.7	673.1	4547.0
Yb	94.42	146.63	245.98	330.41	433.06	2205.77

Table 3: SFH12 Smoothed Data Summary Statistics

These smoothed datasets for each household were then merged into a single dataset, with each smoothed average for the hour based on the smoothing method as a separate column,

ranging from "ya_0" to "ya_23" and "yb_0" to "yb_23". An interesting observation we found was that there are zero missing values aside from missing dates from the Yb smoothing, whereas due to intermittent missing values within the individual household datasets lead to missing hourly moving averages. A heatmap for the missing values is shown below:



Ya Yb

Figure 1: Missing Values

Addressing the missing data issue, we omit the observations for the smoothed values from both the smoothed datasets in order to proceed further with the analysis. We also convert the Household and Weekday as factors with 33 and 7 levels, respectively. Furthermore, we create a binary variable called is_weekend, which is equal to 1 if the Weekday falls on a Saturday or a Sunday, and 0 otherwise. The following would be useful to split the data to observe the effects between them. Lastly, to address the seasonality, we perform Fourier Transformation and create the variable seasonal_sin.

3 Methods

3.1 Statistical methods

In the section we explain all the methods used for data transformation, performing linear regressions, clustering based on the consumption of the households and forecasting for the household using the previous values.

Moving Average For a reduction in the noise in the raw hourly consumption data, and highlighting daily and weekly load patterns, we smoothed the data using Hourly Moving Average (Ya) and Weekly Moving Average (Yb). For the respective moving averages, the window for smoothing was set for ± 30 minutes and ± 2 weeks around each day. The Weekly Moving Average was applied on the data that had already underwent Hourly Moving Average for consistent

results.

$$\text{Hourly Moving Average: } \tilde{Y}_{a_{h,D}} = \frac{1}{w_1} \sum_{i=-k_1}^{k_1} Y_{h+i,\,D}$$

$$\text{Weekly Moving Average: } \tilde{Y}_{b_{h,D}} = \frac{1}{w_2} \sum_{j=-k_2}^{k_2} \tilde{Y}_{a_{h,\,D+14j}}$$

Fourier Transformation To explicitly model the seasonality, we incorporated Fourier transformation as a part of feature engineering for the smoothed data. Here, we used the sin-based seasonal component to capture the seasonality, allowing us to detect the periodic patterns over the year:

$$S_D = \sin(\frac{D - Mar21}{365} * 2\pi)$$

Multivariate ANOVA (MANOVA) After the feature engineering to set up our predictors, we used MANOVA as an initial step to determine which of the model choices was a feasible option to use. (Rencher, 1998a; 2025a). From this approach, we aimed to compare using the p-values from the MANOVA and decided based on their significance value, for which we consider the significance if the p-value was less than 0.05.

Model Without Interaction	Model with Interaction
$\tilde{Y}_{a,b} = W + S + H$	$\tilde{Y}_{a,b} = W * S * H$

Table 4: MANOVA Models

The predictors W, H denoted the Weekday and Household, respectively. These have been converted to factor levels for the model. The predictor S denoted the sin-based seasonal component from the Fourier Transformation.

For the MANOVA, we used Wilks' Test with the given formula:

$$\Omega_{Wilk} = \prod_{i=1}^{n} (\frac{1}{1+\lambda_i}) = \det(I + A)^{-1}$$

$$= \frac{\det(E)}{\det(E+H)} \in [0, 1]$$

Multivariate Linear Regression We modeled multivariate linear regression model using the smoothed load hourly load profiles for the households. (Rencher, 1998b; 2025b) The

following dependent variable, is a vector of 24 hourly values corresponding to the respective smoothed datasets, $\tilde{Y}_{a,h}$ and $\tilde{Y}_{b,h}$.

Let $Y_{H,D,h}$ denote the smoothed load at hour h on day D for household H. For each observation (household-day), we defined the vector

$$Y_{H,W} = (Y_{H,D}^{\intercal})_{F_W(D)=W} = \begin{pmatrix} Y_{H,0}^{\intercal} \\ \vdots \\ Y_{H,23}^{\intercal} \end{pmatrix} \in \mathbb{R}^{24}$$

The multivariate linear regression model was thus formulated as:

$$Y_{H,W} = X_{H,W}^{\intercal}\Phi_{H,W} + Z_{H,W}$$

$$\implies \begin{pmatrix} Y_{H,0}^{\intercal} \\ \vdots \\ Y_{H,23}^{\intercal} \end{pmatrix} = \begin{pmatrix} X_{(0)}^{\intercal} \\ \vdots \\ X_{(23)}^{\intercal} \end{pmatrix} \Phi_{H,W} + Z_{H,w}$$

where $X_{H,w}^{\intercal} \in \mathbb{R}^{24}$ is the design matrix containing regressors including intercept and the seasonality predictor, as well as the dummy variable is_weekend, which is equal to 1 if the weekday factor is equal to a Saturday or a Sunday, which would be useful to estimate the effect weekends have and vice versa on the load profiles. The regressor matrix $\Phi_{H,W} \in \mathbb{R}^{3\times 24}$ captures the effect of each regressor at each hour for each household and whether it falls on a weekend or otherwise, and $Z_{H,W} \in \mathbb{R}^{24}$ is the error vector. As an alternative to MANOVA, the Akaike Information Criterion (AIC) for each model and smoothing method was calculated. (James et al., 2013a)

The study used the simple model to determine the OLS coefficients, that is, we did not use interaction model for the reason explained in Section 4. Furthermore, the OLS model with the interaction terms $\tilde{Y}_{a,b} \sim W * H * S$ was also determined using the framework and the AIC was calculated for comparison.

The OLS estimator for the coefficients was given by:

$$\hat{\Phi}_{H,W} = (X_{H,D}^{\intercal}X_{H,D})^{-1}X_{H,D}^{\intercal}Y_{H,W}$$

Each estimated coefficient vector $\hat{\Phi}_{H,W} = (\hat{\phi}_{H,W,0}, \cdots, \hat{\phi}_{H,W,23})$ summarized the influence

of regressors on the hourly load pattern. These coefficient vectors were subsequently used for clustering analysis to identify similar load patterns within the households.

Akaike Information Criterion

$$AIC = 2k - 2\ln \hat{L},$$

where k is the number of parameters in the model and $\hat{L}$ is the model's maximized likelihood. The Akaike Information Criterion was used for the OLS models to compare the better choice of the model for the smoothed data, and we calculated it for the 30 households selected at random, as well as for each of the randomly selected households.

Hierarchical Clustering In order to identify groups of households with similar load profiles, we clustered our households via centroid-based Hierarchical Clustering using the scaled OLS coefficients as features (James et al., 2013b). It is mathematically represented as follows:

$$d_{ij} = |x_i - x_j|^2 = \sqrt{\sum_{k=1}^{p} (x_{ik} - x_{jk})^2}$$

where d_{ij} is the Euclidean distance between the observations i, j and p is the number of features.

We then used Ward's Linkage Criterion for agglomeration:

$$\Delta(K_1, K_2) = \frac{|K_1| \cdot |K_2|}{|K_1| + |K_2|} |\bar{x}_{K_1} - \bar{x}_{K_2}|^2$$

where K_1, K_2 are the arbitrary clusters, $|K_1|, |K_2|$ the sizes of the clusters and $\bar{x}_{K_1}, \bar{x}_{K_2}$ their respective centroid vectors.

The following approach was then independently repeated for the data after splitting for weekend (Saturdays - Sundays) and weekday (Monday - Friday) using the is_weekend dummy variable, allowing to detect the distinct consumption patterns for both smoothed load profiles.

Mean Load Profile After clustering the households based on their scaled OLS coefficient vectors, we constructed the representative load profiles for each cluster, also known as Mean Load Profiles. Such profiles help summarize the typical electricity consumption pattern for each cluster at each hour of the day. (Andersen et al., 2013)

For each cluster $K \in (1, X_W)$ for each day D and each hour h, the mean load profile for each smoothing method was calculated as

$$\hat{Y}_{D,K}^{(h)} = \frac{1}{|C_{K,F_W(D)}|} \sum_{\hat{\Phi}_{H,F_W(D)} \in C_{K,F_W(D)}} (1, S_D)\hat{\phi}_{H,F_W(D),h}$$

As these capture the characteristic daily consumption pattern of the households within each cluster, it is used both for interpretation of cluster behavior and as an exogenous input in subsequent forecasting models.

Simultaneously, we assessed the variability within the clusters and calculated the Mean Prediction Error with the given formula:

$$\hat{\sigma}_{D,K}^{(h)} = \frac{1}{|C_{K,F_W(D)}|} \sum_{H:\hat{\Phi}_{H,F_W(D)} \in C_{K,F_W(D)}} (Y_{H,D,h} - Y_{D,K}^{(h)})^2$$

This helps to understand the dispersion of individual household load profiles around the cluster-based Mean Load Profile. A confidence interval around the Mean Load Profile using the Mean Prediction Error $\hat{Y}_{D,K}^{(h)} \pm 3 \cdot \hat{\sigma}_{D,K}^{(h)}$ is subsequently generated to quantify the uncertainty for the forecasting process. (2025c)

Forecasting With the calculation of the Mean Load Profiles based on the clustering, we now proceed with the forecasting of the electricity consumption for the remaining three households. (2025d) Here, we have the following vectors that denote the incumbent values (state) and the observed values, as $X_t = \begin{pmatrix} X_{t,1} \\ X_{t,2} \\ X_{t,3} \end{pmatrix}$, $Z_t = \begin{pmatrix} Z_{t,1} \\ Z_{t,2} \\ Z_{t,3} \end{pmatrix}$ respectively. With the available time indexing $f_T(t) := \{1, \cdots, 365 \cdot 24\} \in t \to f_T(t) = (D(t), h(t)) \in \{1, \cdots, 365\} \times \{1, \cdots, 24\}$ for each corresponding day and hour to the time index t, we have for each household H the state equation as an AR(1) model: $X_{t,H} = \theta_X X_{t-1,H} + \theta_v v_{t,H} + E_{t,H}^X$, with the process noise $E_{t,H}^X \sim WN(0, Q_{t,H})$, where $Q_{t,H} = (\hat{\sigma}_{D(t),K}^{(h(t))})^2 + \sigma_X^2$, and the noise having a Gaussian distribution.

The variable $v_{t,H} = \hat{Y}_{D(t),K}^{h(t)}$ represents the exogenous Mean Load Profile calculated from the classification for each smoothing method, that is if the coefficient matrix $\Theta_{H,f_W(D(t))}$ belongs to the respective cluster $C_{K,f_W(D(t))}$, determined by the consumption pattern of the household. The variable is omitted in the case of determining without any individual load profile in the

state model.

The observation equation for each household follows from the state equation with: $Z_{t,H} = \theta_Z X_{t,H} + E^Z_{t,H}$, and the observation model having the noise $E^Z_{t,H} \sim N(0, \sigma^2_Z)$.

For the forecasting, we employ Kalman Filter and Particle Filter for the households, where both the methods use parameters. The parameters can be represented as a matrix, $(\theta_X, \theta_v, \theta_Z, \sigma^2_X, \sigma^2_Z)$, and to ensure their robustness and consistency, we start with the following set: $(1, 1, 1, 0.01, 0.01)$ and optimize them further using L-BFGS-B (Byrd et al., 1995) as the optimization algorithm to get the optimal set of parameters.

Kalman Filter Developed by Kálmán in 1960, this approach models the latent consumption state and the observed values jointly for all the households, with the assumption that both the processes for the state and observation are linear in nature with a Gaussian noise. Because of these assumptions, the method has a low computational cost, as it depends on the closed system of equations that will be highlighted further in the section. (Petris et al., 2009a; 2025d)

The general State equation, showing the transition, is represented as follows:

$$X_t = \Theta_X X_{t-1} + \Theta_v v_t + E^X_t,\ E^X_t \sim WN_p(0_p, Q_t),\ Q_t \in \mathbb{R}^{p\times p}$$

where $\Theta_X \in \mathbb{R}^{p\times p}$ is the state transition matrix, $\Theta_v \in \mathbb{R}^{p\times q}$ being the exogenous input matrix, and E^X_t is the process noise with the covariance $Q_t \in \mathbb{R}^{p\times p}$.

We also have the general Observation equation:

$$Z_t = \Theta_Z X_t + E^Z_t,\ E^Z_t \sim WN_m(0_m, R_t),\ R_t \in \mathbb{R}^{m\times m}$$

with $\Theta_Z \in \mathbb{R}^{m\times p}$ as the observation matrix, and E^Z_t the observation noise with covariance R_t.

In the Kalman Filter, the state $X_0 \sim WN_p(c_0, \rho_0)$ is assumed to have a Gaussian distribution, as the initialization process. The next state $X_t|Z_{t-1}$ with it's uncertainty is denoted by

$$a_t = \Theta_X Z_{t-1} + \Theta_v v_t,\ p_t = \Theta_X C_{t-1} \Theta^\intercal_X + Q_t$$

And as observation depends on the state, so does the prediction $Z_t|Z_{t-1}$ for the next observation:

$$b_t = \Theta_Z a_t,\ S_t = \Theta_Z p_t \Theta^\intercal_Z + R_t$$

As the actual observation is made available as a result of the prediction step, the state estimator $X_t|Z_{t-1} \sim WN_p(c_t, \rho_t)$ is therefore updated as a result:

$$c_t = a_t + K_t(Z_t - \Theta_Z a_t) = a_t + p_t \Theta_Z^\intercal S_t^{-1}(Z_t - \Theta_Z a_t)$$

with $\rho_t = (\mathbb{I}_{p\times p} - K_t\Theta_Z) \cdot p_t = p_t - p_t\Theta_Z^\intercal S_t^{-1}\Theta_Z p_t$, where $K_t = p_t\Theta_Z^\intercal S_t^{-1}$ is the Kalman Gain, and $\mathbb{I}_{p\times p}$ is the identity matrix. Therefore, any updates to the predictions involve the use of Kalman Gain.

For this report, we let $m = p = 3$, & $q = 1$. Our multivariate coefficients and the noise are therefore denoted as $\Theta_X = \theta_X\mathbb{I}_{3\times3}, \Theta_v = \theta_v\mathbb{I}_{3\times3}, \Theta_Z = \theta_Z\mathbb{I}_{3\times3}$, $Q_t = diag(Q_{t,1}, Q_{t,2}, Q_{t,3}), R_t = \sigma_Z^2\mathbb{I}_{3\times3}$.

Particle Filter The Particle Filter is a flexible, simulation-based approach for the forecast, particularly suited for non-linear state models with a non-Gaussian noise. Compared to Kalman Filter, which relies on closed-form updates under the linear Gaussian assumptions, the Particle Filter uses a set of random samples, called particles, and their associated weights as a measure to generate forecasts. (Petris et al., 2009b; 2025e)

For each household $k \in \{1, 2, 3\}$, we initialized the process by drawing the i.i.d sample of particles $x_0^{(1)}, \cdots, x_0^{(M)}$ from $N(c_0, \rho_0)$ and calculated their mean for each i:

$$\bar{x}_0 = \frac{1}{n}\sum_{i=1}^{M} x_0^{(i)}$$

For the prediction process, at each time t, we then drew an i.i.d sample of errors $e_0^{(1)}, \cdots, e_0^{(M)}$ from $N(0, \theta_t)$ and propagated the particles according to the model:

$$\tilde{x}_t^{(m)} = \theta_x\bar{x}_{t-1} + \theta_v v_t + e_t^{(m)}$$

For $m = 1, \cdots, M$, we calculated the weights according to the likelihood of the observed values z_t given each particle $\tilde{w}_t^{(m)} = f_{w(\theta_t\tilde{x}_t^{(m)}, \sigma_t^2)}(z_t)$. The weights were the normalized by dividing them with their overall sum

$$w_t^{(m)} = \frac{\tilde{w}_t^{(m)}}{\sum_{m=1}^{M}\tilde{w}_t^{(m)}}$$

After weighting, we then performed resampling to the particles. We started with drawing i.i.d samples $x_t^{(1)}, \cdots, x_t^{(M)}$ from random variable X with the discrete distribution given by

$P(X = \tilde{x}_t^{(m)}) = w_t^{(m)}$ for $m = 1, \cdots, M$ that we calculated from the normalized weights. Then the empirical distribution over the state at time t was given by

$$\Pi_t(x_t) = \frac{1}{n} \sum_{m=1}^{M} \delta_{x_t^{(m)}}(x_t)$$

where δ_x denoted the respective Dirac function.

We then had the mean for the state and predicted observations as follows:

$$\bar{x}_t = \frac{1}{n} \sum_{m=1}^{M} x_t^{(m)}$$

$$\bar{z}_t = \theta_z \bar{x}_t = \frac{1}{n} \sum_{m=1}^{M} \theta_z x_t^{(m)}$$

The empirical variance of the predicted observation was therefore

$$\hat{\sigma}_{z,t}^2 = \frac{1}{n-1} \sum_{m=1}^{M} (\theta_z x_t^{(m)} - \bar{z}_t)^2,$$

, with the standardized prediction error $\frac{z_t - \bar{z}_t}{\hat{\sigma}_{z,t}^2} \sim N(0, 1)$

We then assessed the quality of our forecast for each case and household by performing diagnostics, including Box and Shapiro tests (Petris et al., 2009c), and a comparative analysis of the Root Mean Squared Error.

Root Mean Squared Error The following method is used to evaluate the quality of forecasts, by measuring the average magnitude of the errors between the predicted and the observed values, with lower RMSE values indicating better predictive performance (James et al., 2013c). It is formulated as the following:

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^{n} (x_t - z_t)^2}$$

where x_t denotes the actual value and z_t denotes the observed value at time t.

For the diagnostic tests, below is the table listing the null and alternate hypotheses:

	Null Hypothesis	Alternate Hypothesis
Box Test	Independent errors	Autocorrelation present in errors
Shapiro-Wilk Test	Errors Normally distributed	Errors not Normally distributed

Table 5: Test statistics

3.2 Software

R was used mainly for the project, including the OLS modeling, clustering, Mean Load Profile calculation and forecasting. We used the following libraries: car, systemfit, cluster, matrixStats, lme4, and ggplot2 (Wickham, 2016). (R Core Team, 2025)

In addition to R, the analysis for the missing data, and the smoothing of the load profiles for the households was done in Python using Seaborn, Matplotlib and Pandas. (Van Rossum and Drake, 2009).

We also sparingly used the GitHub Copilot (model GPT-4.1) for technical debugging. OpenAI, 2025

4 Empirical Results

After obtaining the smoothed data, we then identified whether a simple linear model, or a model with interaction was more feasible. Initially we used Multivariate ANOVA, and observed that in the case of Ya smoothing, the variable household is highly insignificant, but the rest of the variables have a very inflated significance value, for both the models. We observe the same MANOVA results for Yb smoothing (table in the Appendix), with a difference of household variable being highly significant. These results were inconclusive and we moved to compare the Akaike Information Criterion for each OLS model.

	Df	Wilks	approx F	num. Df	den. DF	Pr(>F)
household	29	0.98310	0.256	696	174668	1
day_of_week	6	0.29526	88.792	144	55084	<2.2e-16
seasonal_sin	1	0.68517	180.428	24	9424	<2.2e-16
Residuals	9447					

Table 6: MANOVA (Ya, No Interaction)

From the AIC calculated from the OLS Models incorporating all the 30 selected households, we had compiled the results in the table below.

Here, the AIC showed that the models with interactions were a better choice to work with, with a significant difference between their simple counterparts. However, as the results were for all the households combined, we also investigated whether the effect remained for each household separately, and the AIC results were interesting. Here, because the AIC between the models was so insignificant, none of the houses required an interaction model, whereas for Yb smoothing, we

	Df	Wilks	approx F	num. Df	den. DF	Pr(>F)
household	29	0.98040	0.257	696	167574	1
day_of_week	6	0.28574	87.726	144	52485	<2.2e-16
seasonal_sin	1	0.0.67484	181.511	24	9041	<2.2e-16
household:day_of_week	174	0.96264	0.083	4176	215198	1
household:seasonal_sin	29	0.98959	0.136	696	167574	1
day_of_week:seasonal_sin	6	0.48040	49.409	144	52485	<2.2e-16
household:day_of_week:seasonal_sin	174	0.089	5.906	4176	215198	1
Residuals	9064					

Table 7: MANOVA (Ya, Interaction)

OLS Model	AIC
Ya, No Interaction	3420435
Ya, Interaction	3101706
Yb, No Interaction	3412379
Yb, Interaction	3101664

Table 8: Akaike Information Criterion for OLS Models

observe that a majority of houses require interaction in their model as the AIC difference was more significant. For consistency, and that the simple model was less computationally expensive to build the coefficient vectors for clustering, we decided with using the simple model for the rest of the analysis.

Smoothing Method	Houses
Ya	0
Yb	22

Table 9: Number of Houses that require Interaction Model

As we obtained the coefficients for the OLS models without interaction for Ya and Yb smoothing, we then classified the selected households, as well as the three remaining households in their respective clusters. Here, we had 72 coefficients for 30 households initially before splitting them based on the weekend dummy variable to observe the cluster effect, effectively having 48 coefficients for each respective case. We investigated the optimal k for the clustering using Elbow and Silhouette Plots for both the smoothed OLS coefficient matrices.

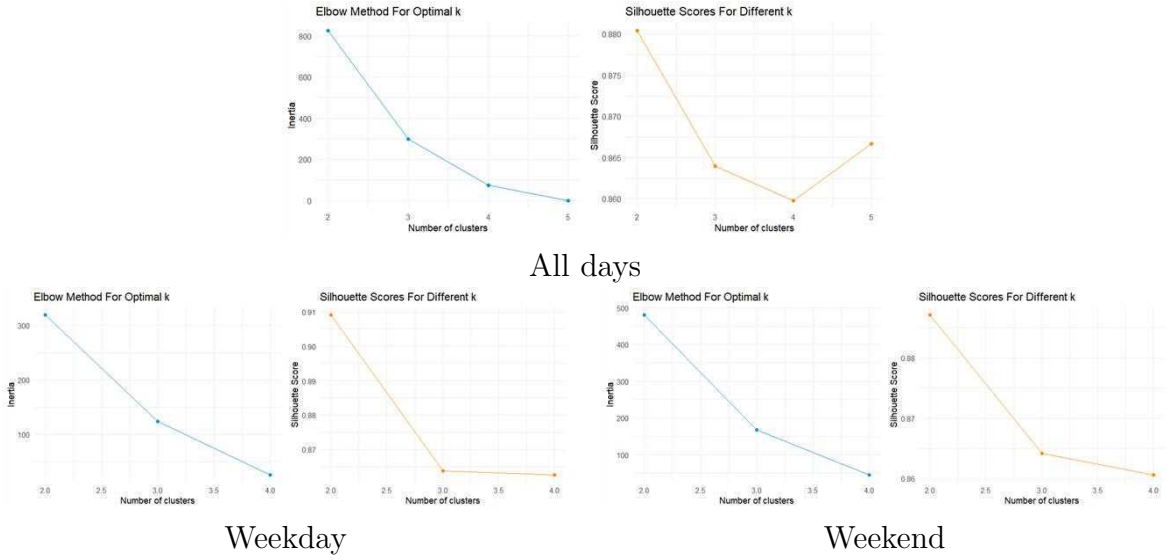


Figure 2: Ya: Elbow and Silhouette Plots

In the case for Ya, we observed that there is a sharp drop in the inertia from the elbow method, for the aggregated version, as well as the coefficients after splitting for the weekend, at $k = 3$.

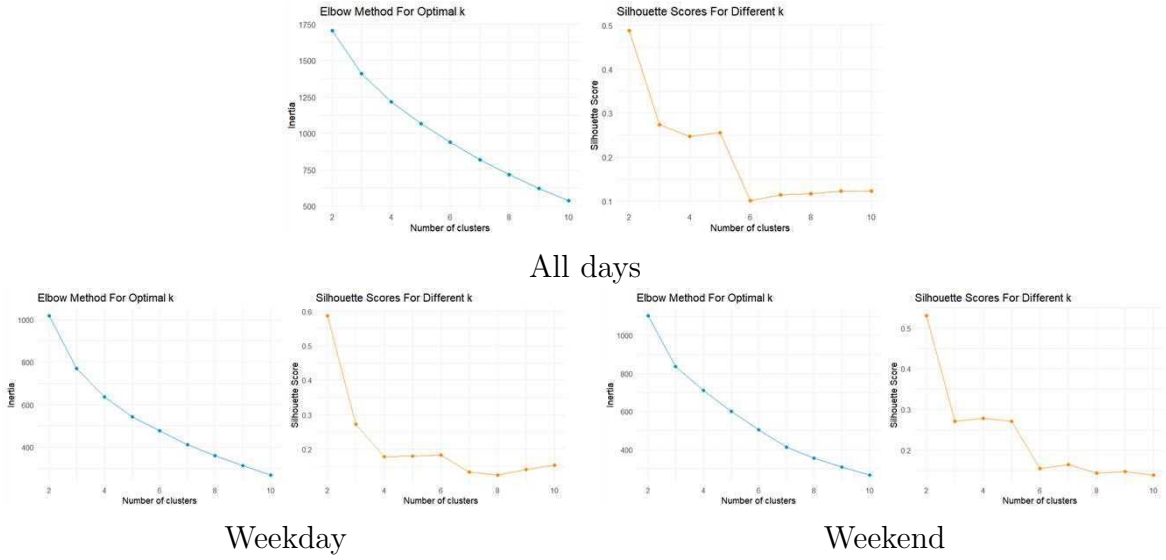
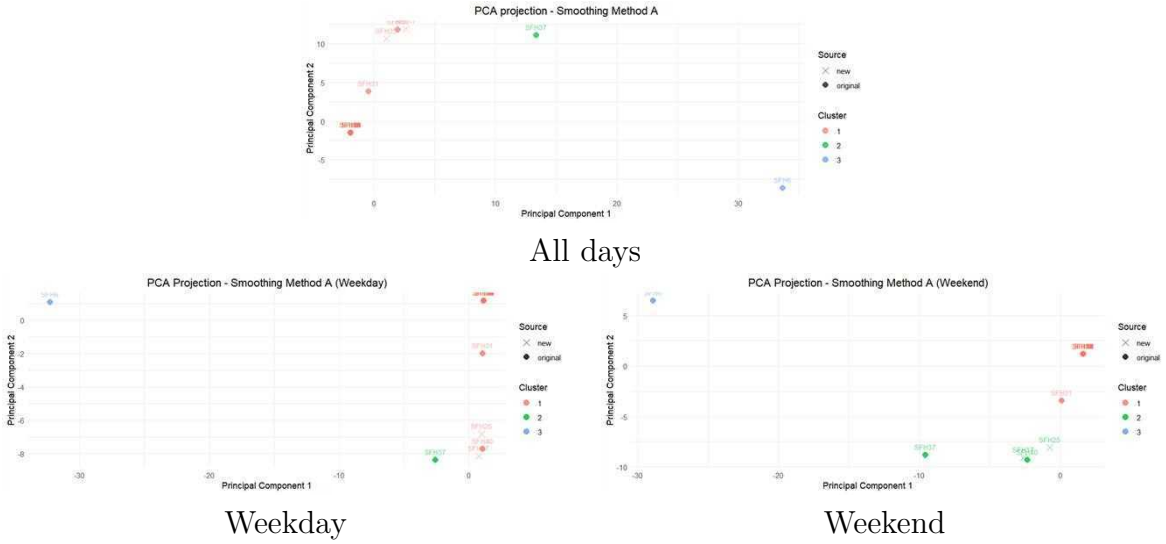


Figure 3: Yb: Elbow and Silhouette Plots

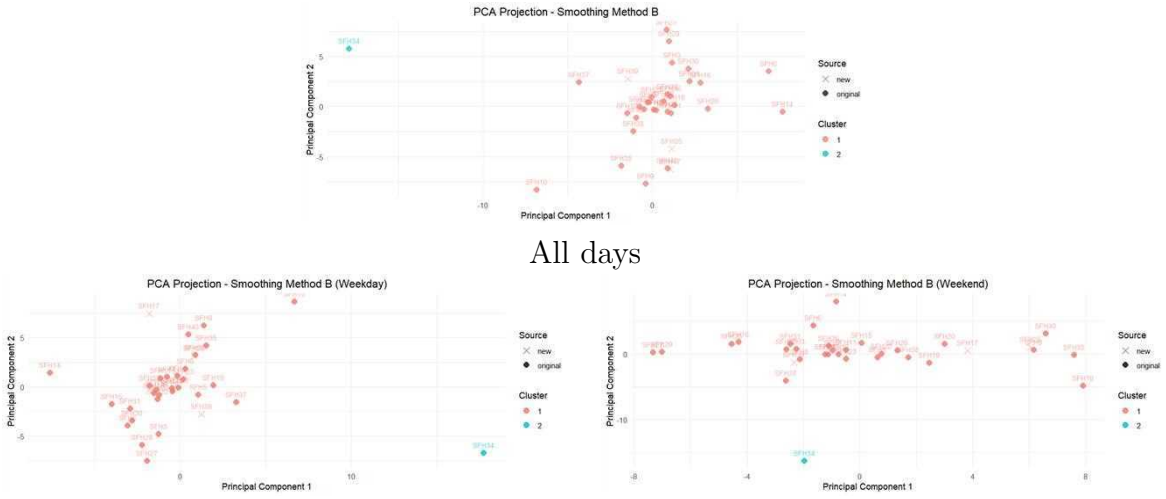
For Yb, we observed that the Elbow method did not explain well the optimal k, so we looked at the Silhouette Score plot for each case, and with the highest score, we determined $k = 2$ as the optimal number for clustering. For the remaining three households, the optimal k value remains the same for each option.



All days

Weekday Weekend

Figure 4: Ya: PCA Projection Plots



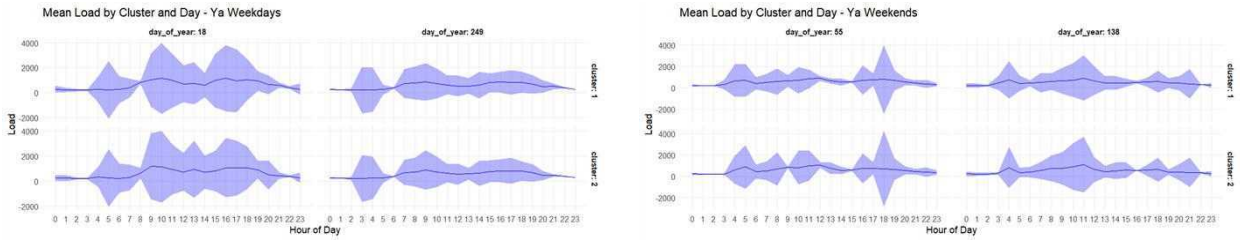
All days

Weekday Weekend

Figure 5: Yb: PCA Projection Plots

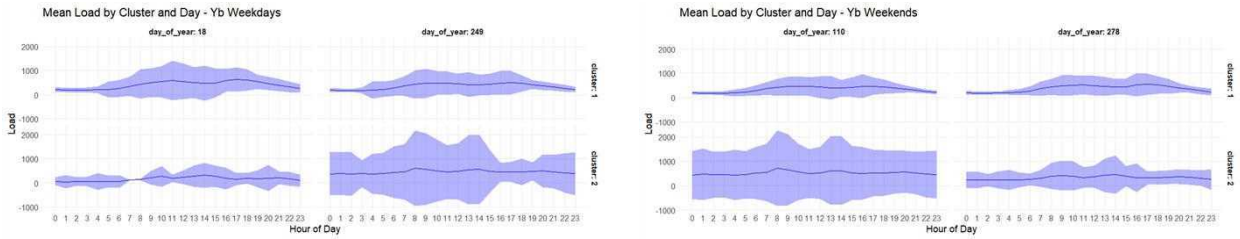
From the selection, we observed that the remaining households were "SFH17", "SFH25" and "SFH39". Together with the 30 selected households, we plotted the PCA Projection graphs to see the clustering results, and the remaining households remain in Cluster 1 with the coefficients obtained from the OLS modeling with the Yb smoothed data, and there is no effect on the clusters from the weekend/weekday effect. However, a noticeable observation we found was how the households 17 and 25 shifted from Cluster 1 to the much smaller Cluster 2 on the weekends, whereas on the weekdays all the houses remained in the same cluster. This shows that with the hourly smoothing, the households exhibit the consumption pattern quite similar to household SFH37.

With the clusters obtained from the classification, we proceeded to calculate the mean load profile and the mean prediction error (as a proxy for confidence interval) for each household based on their cluster.



Weekday Weekend

Figure 6: Mean Load Profile Graphs (Ya Smoothing)



Weekday Weekend

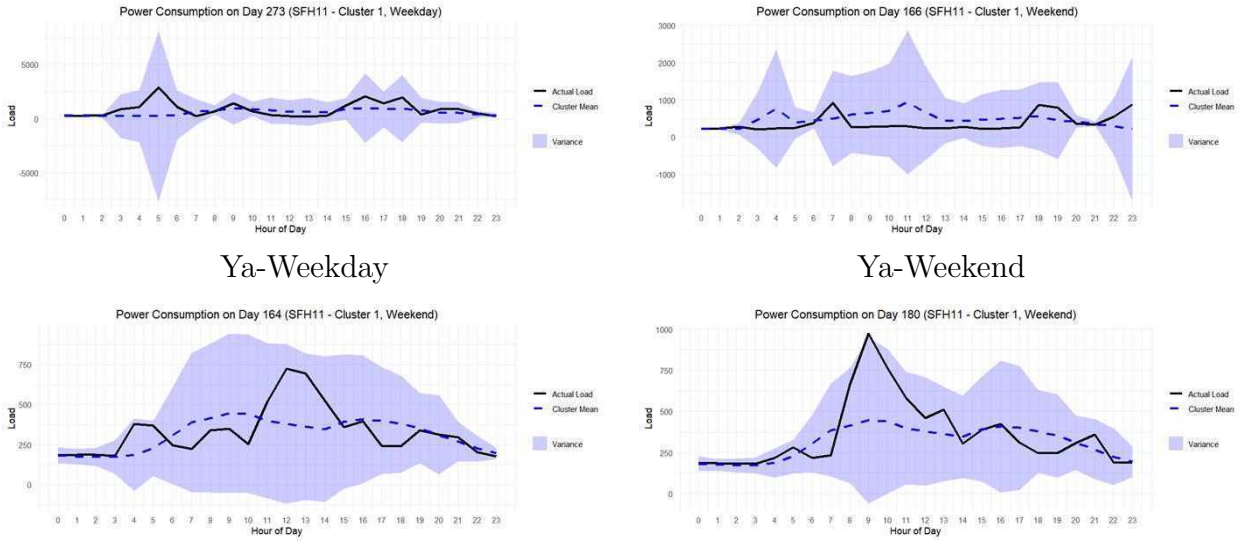
Figure 7: Mean Load Profile Graphs (Yb Smoothing)

For the weekday and weekend selected at random for the clusters obtained from the smoothed datasets, there were some interesting insights from the Mean Load Profiles. From all the graphs, we obtain that generally, all the households consumed more electricity on average at around 08:00 to 10:00, and the peaks spanning between 18:00 and 20:00 also suggest a higher electricity usage, with the least usage observed between 00:00 to 03:00 and 22:00 to the end of the daily cycle, providing us a pattern of usage in general.

Compared to Yb Smoothing, we observe that the mean prediction error has a higher fluctuation rate in the case of Ya Smoothing, with higher peaks observed for a short amount of time, showing varied usage of electricity among households in the clusters. The phenomenon was more noticeable with Cluster 2 with increased fluctuation in the mean prediction error for the same point in time for the mean load for Cluster 1.

Conversely, we noticed that the mean prediction error was the lowest for Cluster 1 Mean Load Profiles from Yb Smoothing, which could imply more confident forecast values. Depending on the day, Cluster 2 displayed much higher (or lower), but a mostly uniform mean prediction

error around the mean load profile, corresponding to the respective peaks and troughs, which is explained by the fact we observed only one household (SFH34) belonged to Cluster 2 when smoothed using Yb.



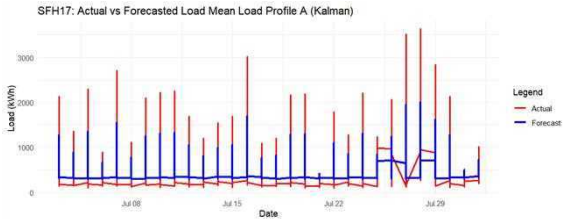
Ya-Weekday Ya-Weekend

Yb-Weekday Yb-Weekend

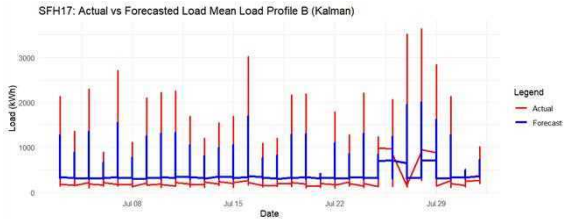
Figure 8: Comparison Graphs with Actual Load

We also compared our mean load profiles for a randomly selected cluster for each case to the actual load of a household belonging to the said cluster, to observe whether the actual use falls within the mean prediction error. For a good measure, we chose Household SFH11 (for having no missing values of either sort) from Cluster 1, and compared the daily load profile to the respective mean load profiles of the days chosen at random. As expected, we observed the actual load not only mostly remained around the mean load profile, but was also contained within the error interval, confirming our results thus far.

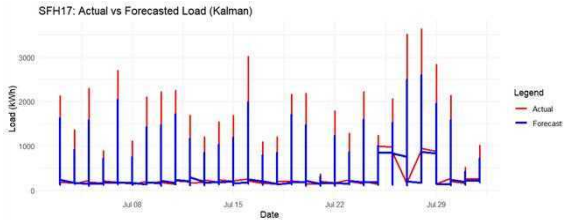
For the forecasts, we have limited the graphs to show the data up to August 1, 2019 for better visualization and consistency. With the estimated parameters used, obtained from L-BFGS-B optimization algorithm using the optim function, we see the forecasts remaining close to the actual values, but the variation and the uncertainty between the forecasted and the actual values are less observed when the forecasting is done with incorporating the Mean Load Profiles using data from the smoothing methods and the clustering based on their properties.



Mean Load Profile A



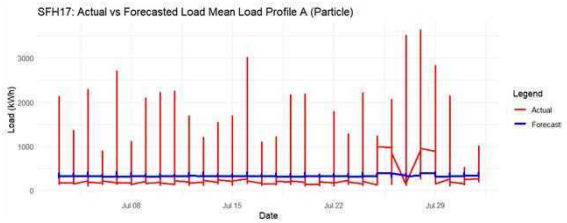
Mean Load Profile B



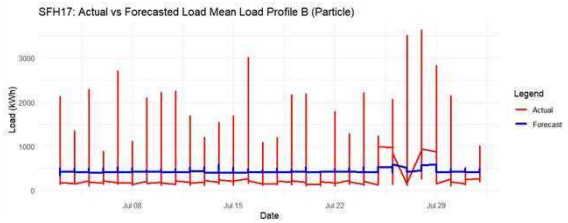
No Individual Load Profile

Figure 9: SFH17 Forecast: Kalman Filter

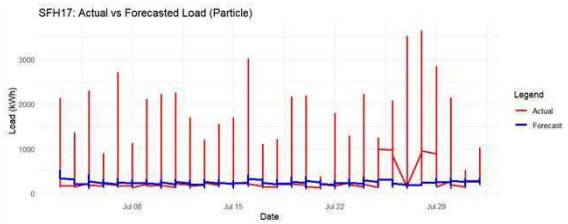
This is even more noticeable with Particle Filter, which was much more computationally expensive than Kalman Filter. The forecast values remain even closer to the actual values for all the cases, but the uncertainty was much less as seen by the very short peaks. With the exogenous input of the Mean Load Profiles we calculated from before, we observed that similar to Kalman Filter results, we obtained even much less variance in the forecasted values, indicating that for the past values, the forecasts for the Particle Filter remained much more stable than what we obtained from Kalman Filter.



Mean Load Profile A



Mean Load Profile B



No Individual Load Profile

Figure 10: SFH17 Forecast: Particle Filter

The results follow the same for Households 25 and 39, so we have put their following load graphs in the Appendix section.

Consequently, we wanted to compare the feasible forecasting method for the households, but the results were an interesting mix. Generally, we observed that the RMSE for the forecasts with the exogenous Mean Load Profiles were lower than using just the actual load profile data for the households to forecast the values. For the household 17, we cannot exactly conclude the better forecasting model, as while the RMSE was observed lower by forecasting with Kalman Filter without the load profile and even with Mean Load Profile A, the RMSE with the Mean Load Profile B was higher than with the Particle Filter. We observed for household 25, that Kalman Filter was a better choice to forecast values, with the RMSE values being lower than that from the Particle Filter for all the profiles. For the last household, Household 39, we see the opposite, that is not only the lowest RMSE values recorded from the households, but the Particle Filter has generated less than Kalman Filter forecast observations.

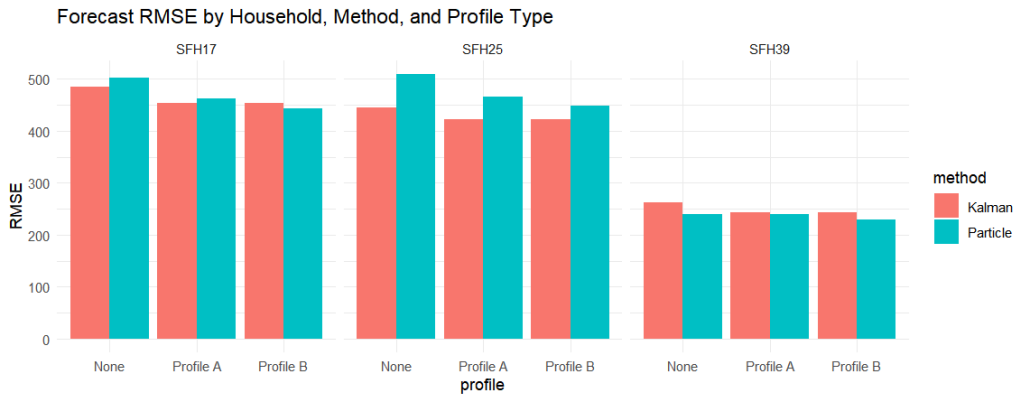


Figure 11: RMSE Comparison

We further conducted the diagnostics for the errors, and we found out that the errors from the Particle Filter did not follow the Normal Distribution, whereas these could be seen with the errors from Kalman Filter, but very loosely. We consequently conducted Box test to check for independence between the errors, and Shapiro-Wilk test for the normality of the errors. With the Box test p-values extremely close to 0 for all instances, the null hypothesis of independent errors was violated and we can conclude the existence of autocorrelation between the errors. Furthermore, the values from Shapiro-Wilk test also appeared to be highly close to 0, well below the critical region, hence we rejected the null hypothesis for the errors to have normal distribution, and therefore concluded the errors did not have a normal distribution. For the values, refer to Appendix Table 12.

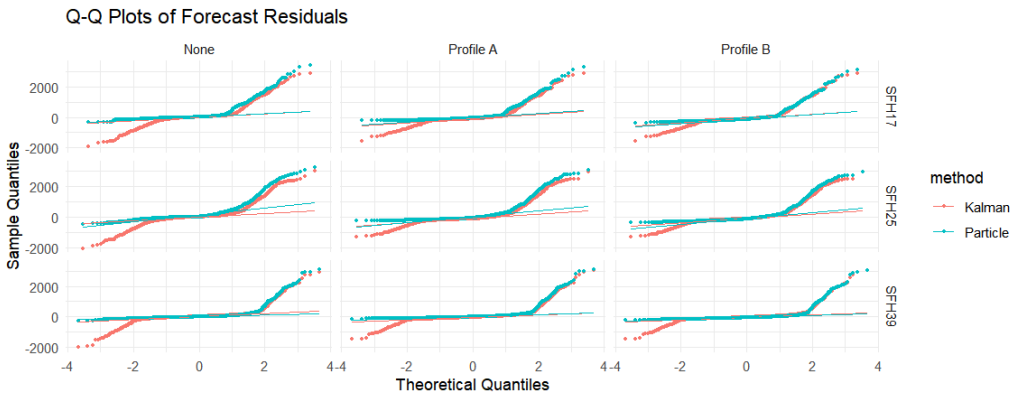


Figure 12: Q-Q Plots for Residual Normality

5 Conclusion

To conclude, we underscored the importance of a robust electricity load forecasting system to enhance the operational efficiency and sustainability within the household energy systems.

Throughout the study, we systematically explored the predictive capabilities of both the Kalman and Particle Filters, utilizing smoothed load profiles and Hierarchical clustering to group households with similar consumption patterns. By integrating the Mean Load Profile as an exogenous feature, we were able to capture the usage patterns, allowing us to interpret the forecasts in a better manner.

The empirical results revealed the Mean Load Profiles generally led to a smaller prediction error, as we observed with the RMSE compared to the analysis relying entirely on the original household data. While Kalman Filter performed better with the assumptions for a linear state-space model and Gaussian White Noise distribution, the Particle Filter seemed as the better choice for the households with more non-linear, non-Gaussian load profiles. However, as seen from the empirical results, the selection of the filter depends on the data of the households. Additionally, as we observed in the beginning, some of the households used in the forecast had the issue of missing dates, therefore contributing to the ambiguity in deciding the more feasible model.

From the residual diagnostics, we observed how errors had autocorrelation present as well as their distribution not being normal. This suggests for further refinements, such as explicitly modeling autocorrelated or heteroscedastic errors, that could improve the forecasting accuracy. Furthermore, we observed the much higher computational cost of Particle Filter, due to its nature, that needs to be taken into consideration.

For potential further improvements to the research, we may obtain the OLS coefficients using the models with interaction, or even a hybrid approach, whereby Ya smoothing uses the simple OLS model for coefficient clustering and Yb smoothing uses interaction OLS model. Another suggestion could be to use more efficient optimization techniques that lead to the best set of parameters for better predictions.

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(2025b), Notes provided for OLS Model dated 22.04.2025.

(2025c), Notes provided for Mean Load Profile dated 13.05.2025.

(2025d), Notes provided for Kalman Filter dated 20.05.2025.

(2025e), Notes provided for Particle Filter dated 27.05.2025.

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6 Appendix

	Df	Wilks	approx F	num. Df	den. DF	Pr(>F)
household	29	0.00813	74.462	696	174668	< 2.2e-16
day_of_week	6	0.71497	22.603	144	55084	<2.2e-16
seasonal_sin	1	0.68071	184.178	24	9424	<2.2e-16
Residuals	9447					

Table 10: MANOVA (Yb, No Interaction)

	Df	Wilks	approx F	num. Df	den. DF	Pr(>F)
household	29	0.00119	105.591	696	167574	< 2.2e-16
day_of_week	6	0.55184	39.296	144	52485	<2.2e-16
seasonal_sin	1	0.57199	281.882	24	9041	<2.2e-16
household:day_of_week	174	0.01652	9.706	4176	215198	<2.2e-16
household:seasonal_sin	29	0.07633	36.408	696	167574	<2.2e-16
day_of_week:seasonal_sin	6	0.83899	11.191	144	52485	<2.2e-16
household:day_of_week:seasonal_sin	174	0.07578	5.906	4176	215198	<2.2e-16
Residuals	9064					

Table 11: MANOVA (Yb, Interaction)

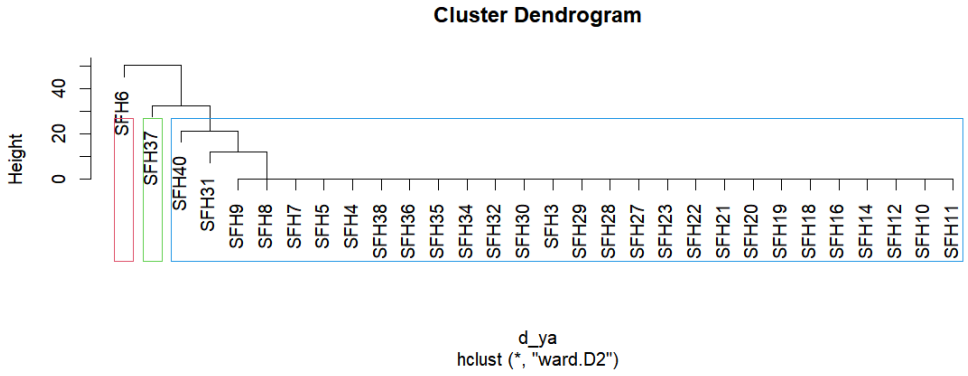


Figure 13: Cluster Dendogram (ya)

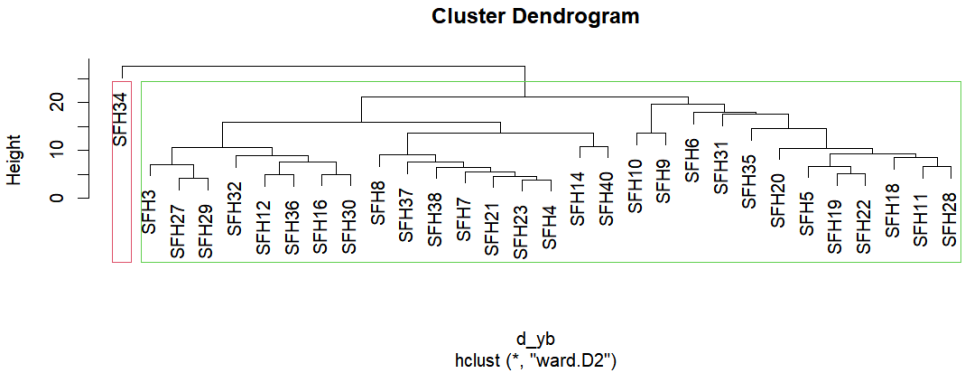
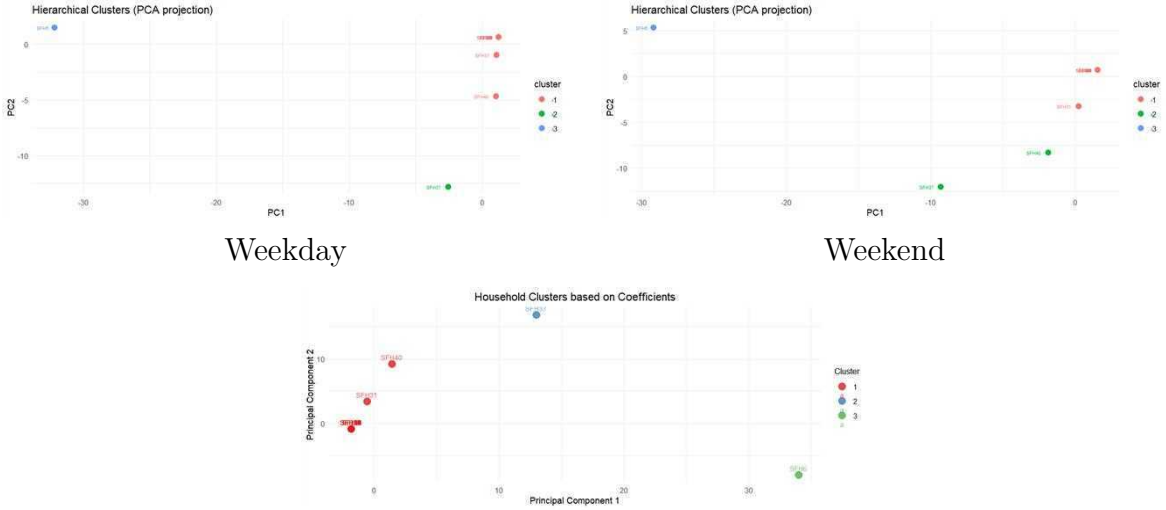


Figure 14: Cluster Dendogram (yb)

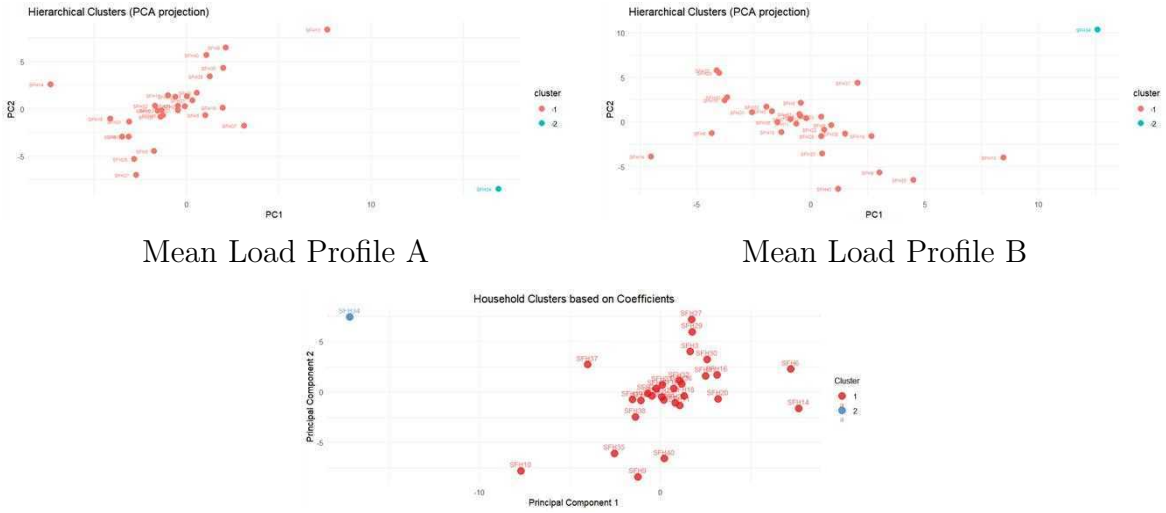


Weekday

Weekend

All days

Figure 15: PCA Projection for Ya - 30 Households only

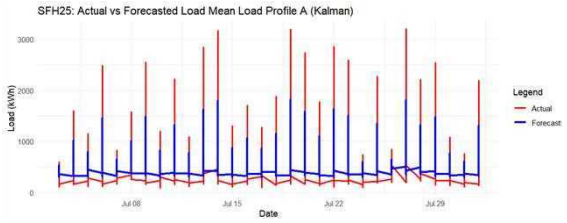


Mean Load Profile A

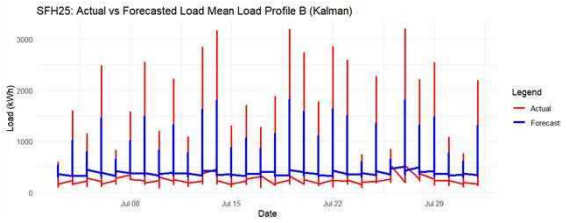
Mean Load Profile B

No Individual Load Profile

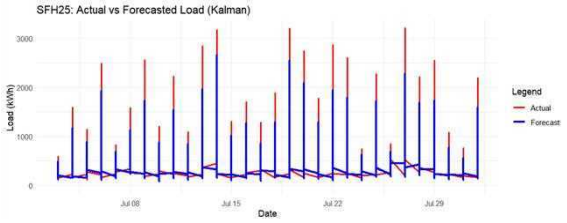
Figure 16: PCA Projection for Yb - 30 Households only



Mean Load Profile A

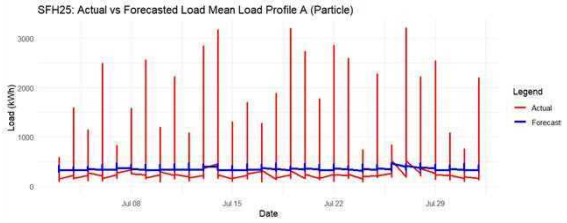


Mean Load Profile B

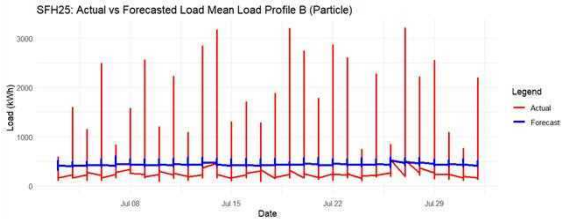


No Individual Load Profile

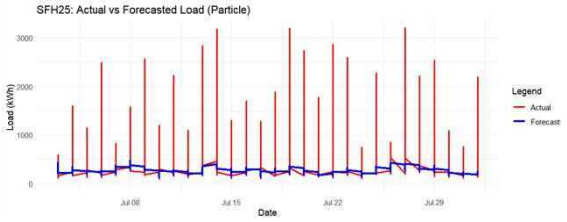
Figure 17: SFH25 Forecast: Kalman Filter



Mean Load Profile A

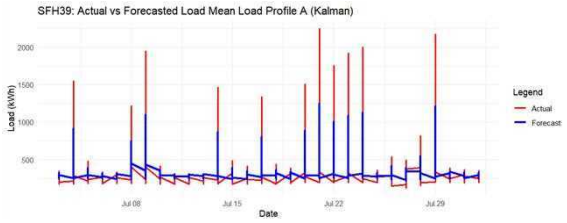


Mean Load Profile B

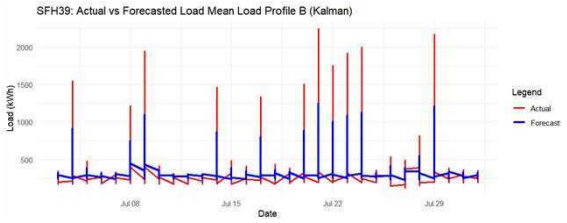


No Individual Load Profile

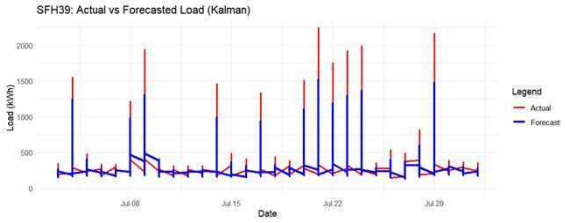
Figure 18: SFH25 Forecast: Particle Filter



Mean Load Profile A

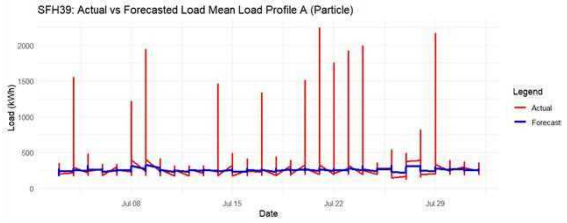


Mean Load Profile B

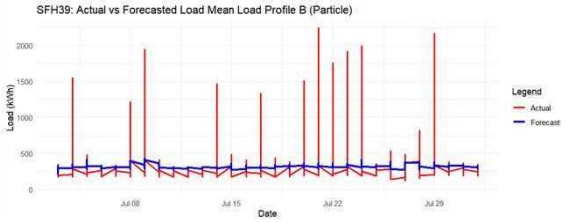


No Individual Load Profile

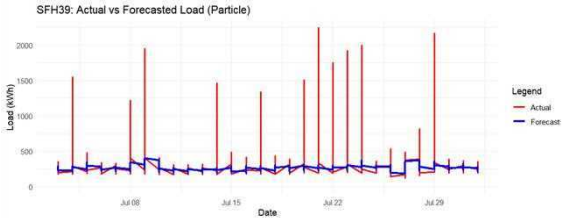
Figure 19: SFH39 Forecast: Kalman Filter



Mean Load Profile A



Mean Load Profile B



No Individual Load Profile

Figure 20: SFH39 Forecast: Particle Filter

Household	Method	Profile	Shapiro_p	Box_p
SFH17	Kalman	None	5.87e-41	0
		Profile A	4.94e-43	0
		Profile B	4.94e-43	0
	Particle	None	1.38e-48	0
		Profile A	1.43e-48	0
		Profile B	5.18e-48	0
SFH25	Kalman	None	2.73e-52	0
		Profile A	1.73e-55	1.38e-11
		Profile B	1.73e-55	1.38e-11
	Particle	None	9.66e-57	0
		Profile A	5.20e-59	0
		Profile B	8.80e-59	0
SFH39	Kalman	None	1.39e-72	0
		Profile A	1.93e-74	0
		Profile B	1.93e-74	0
	Particle	None	5.38e-80	0
		Profile A	5.15e-80	0
		Profile B	2.91e-80	0

Table 12: Table for Shapiro-Wilk and Box Tests